

CodeX

AI Code Review Agent

Intelligent Code Analysis & Quality Assurance using AI

J.N.N Institute of Engineering

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The Challenge Developers Face

The Cost of Manual Code Review

Software developers routinely face hidden bugs, security vulnerabilities, poor code quality, and inefficient algorithms – all while trying to maintain coding standards. Manual code reviews are slow, inconsistent, and resource-intensive, driving up development time, maintenance costs, and security risks.

These compounding issues slow down teams, introduce technical debt, and leave applications exposed to threats – demanding a smarter, automated solution.

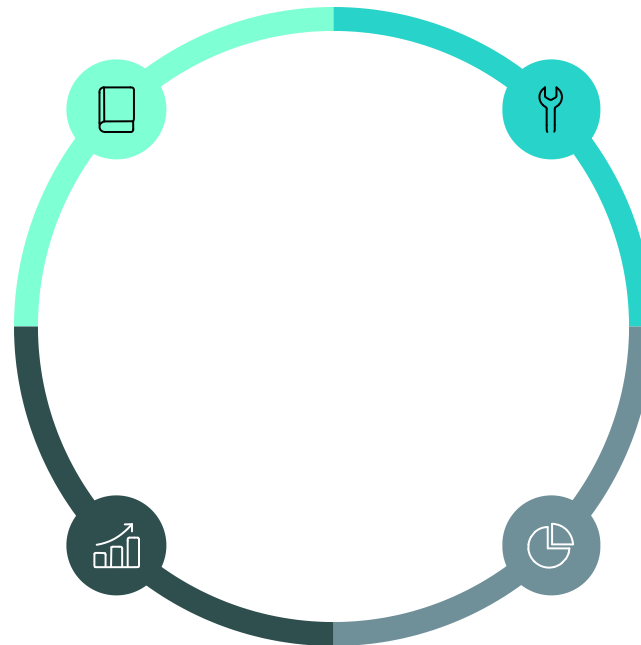
AI Code Review Agent — CodeX

Bug & Error Detection

Automatically identifies hidden bugs, logical errors, and code smells that might slip through manual review processes.

Standards & Quality Scores

Reviews adherence to coding standards and delivers quality scores with actionable, easy-to-understand feedback.



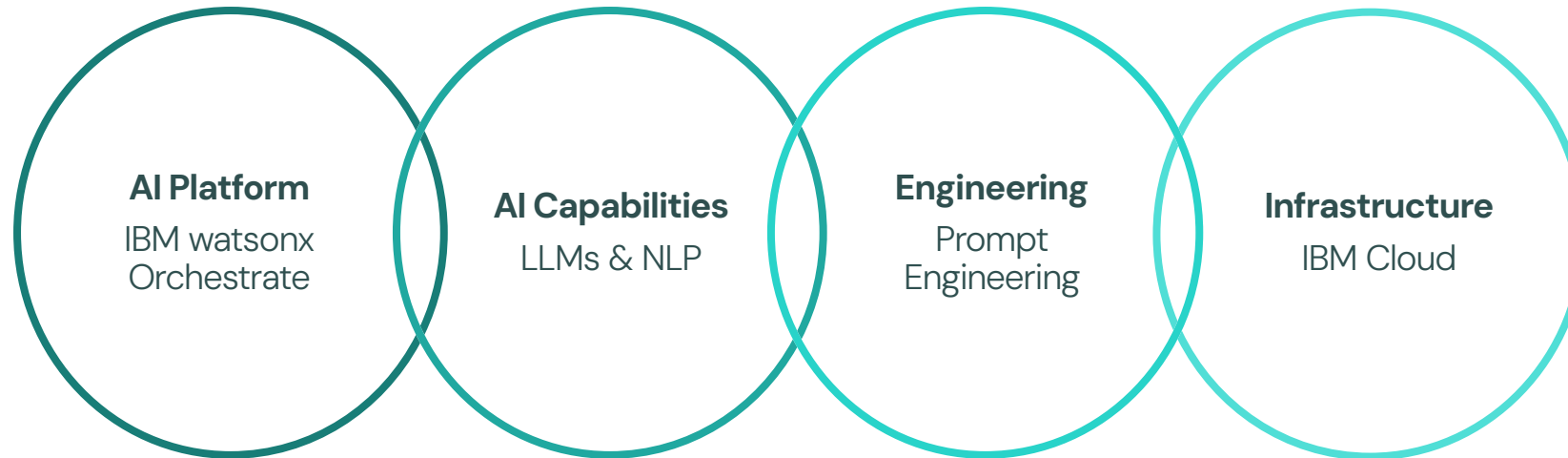
Security Scanning

Detects vulnerabilities and flags security risks before they reach production, protecting applications from exploitation.

Performance Optimization

Suggests refactored code and efficiency improvements to enhance runtime performance and reduce technical debt.

Development Stack



Built on cutting-edge IBM AI infrastructure with support for 8+ programming languages including Java, Python, C, C++, JavaScript, SQL, Go, and Rust.



IBM watsonx Orchestrate

The core AI platform that orchestrates intelligent agents, managing the end-to-end code analysis pipeline seamlessly.



Large Language Models (LLMs) & NLP

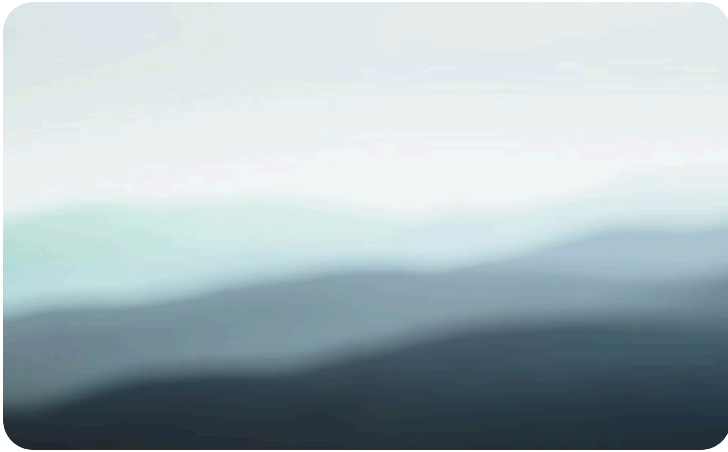
Powering deep code comprehension, pattern recognition, and generation of human-readable improvement suggestions.



Multi-Language Support

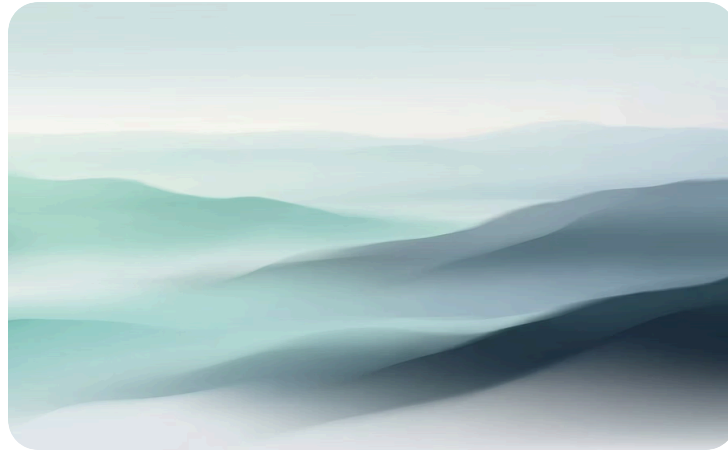
Java, Python, C, C++, JavaScript, SQL, Go, and Rust – CodeX adapts its analysis rules to each language's best practices.

Agent Workflow



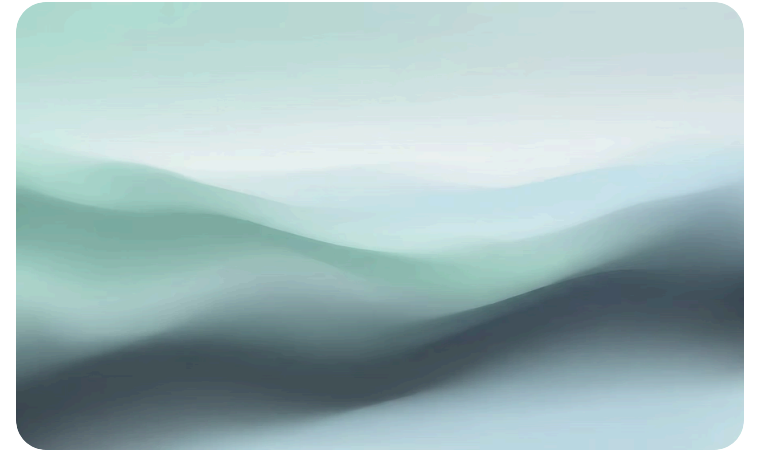
Step 1 – Code Submission

User submits source code. The agent identifies the programming language automatically before proceeding.



Step 2 – AI Analysis

Code is analyzed for bugs, security issues, performance bottlenecks, and code quality. AI generates targeted recommendations.



Step 3 – Report & Refactored Code

Optimized code is suggested and a final review report is displayed, including quality scores and best practice guidance.

Key Features

Comprehensive Automated Code Intelligence

CodeX delivers end-to-end automated review covering every dimension of code quality – from correctness and security to readability and performance.

01

Bug Detection

Automated identification of bugs, logical errors, and code smells across all supported languages with detailed explanations.

03

Performance Optimization & Refactoring

Suggests efficient algorithms and provides refactored code, helping developers adopt best practices immediately.

02

Security Vulnerability Scanning

Deep inspection for known security vulnerabilities with actionable remediation guidance to protect applications.

04

Multi-Language & Standards Support

Supports Java, Python, C, C++, JavaScript, SQL, Go, and Rust – enforcing language-specific coding standards throughout.

Profile

Knowledge

Toolset

Behavior

Channels Preview

Notifications: 1 Dismissed | 1 total

[Legal Notice](#) +

[View all](#)

 Profile

Describe what your agent does so people and other agents know when to use it.

Description*

Helps reviewing Code

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

[What makes a good description?](#) 

Welcome message

Customize the welcome message from the agent on the home screen.

 Running in debug mode

12:43 PM

Hello, Welcome to CodeX

Ready to review your code

Accuracy of generated answers may vary. Please double-check responses.

Refactor this code using
clean coding principles...



Evaluate this code based
on readability,...



Identify syntax, logical,
and runtime errors in thi...



< 1/2 >

Type something...





AI Model : GPT-OSS 120B — O...

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Test agent

live

Deploy

Build with AI

Talk to agent

Running in debug mode

```
import java.sql.*;
import java.util.*;
import java.io.*;
public class EmployeeManagement {
    static Connection con = null;
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) throws Exception {
        Class.forName("com.mysql.jdbc.Driver");
        con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/company",
            "root",
            "root123"); // Hardcoded credentials
        while (true) {
            System.out.println("\n===== Employee Management =====");
            System.out.println("1.Add Employee");
            System.out.println("2.Search Employee");
            System.out.println("3.Delete Employee");
            System.out.println("4.List Employees");
            System.out.println("5.Calculate Salary");
            System.out.println("6.Exit");
            int choice = sc.nextInt();
            switch (choice) {
                case 1:
                    addEmployee();
```

[Save as test](#)

Type something...



AI Model : GPT-OSS 120B — O...

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AI

Running in debug mode

Codex 12:44 PM

EmployeeManagement – Code Review Summary

Area	Rating (1-5)	Comments
Correctness	3	The program compiles and runs, but several logic bugs and unsafe behaviours may cause runtime failures or incorrect results.
Security	2	Hard-coded DB credentials, SQL injection, unchecked input, and unsafe file handling.
Performance	2	Unnecessary nested loops, duplicated code, and inefficient I/O.
Maintainability / Readability	2	Mixed responsibilities, magic numbers, poor naming, and a lack of resource cleanup.
Overall	2.5	Works for a demo, but needs major refactoring before production use.

Below is a detailed walkthrough of the most important problems, why they matter, and how to fix them. Wherever possible I also provide a **ref** **red code snippet** that addresses the issue while

Save as test

Type something...

RESULTS & BENEFITS

Results & Benefits



Faster, Better Reviews

Faster code reviews, improved software quality, reduced debugging time, and enhanced application security – all delivered automatically.



Developer Productivity

Saves time, reduces manual effort, encourages clean coding habits, and improves overall software reliability and maintainability.



Increased Developer Confidence

Developers gain clear, actionable insights into their code, building confidence, better skills, and a culture of quality-first engineering.



CONCLUSION & FUTURE SCOPE

Conclusion & Future Scope

Conclusion

CodeX streamlines the software development lifecycle by providing automated, intelligent, and reliable code reviews — helping developers produce secure, efficient, and maintainable code while reducing manual effort.

Near-Term Roadmap

GitHub, GitLab & Bitbucket integration, CI/CD pipeline support, and real-time IDE plugins to embed CodeX directly into developer workflows.

Advanced Future Features

Team collaboration tools, AI-powered code auto-fix, custom coding standards enforcement, and advanced security compliance checks.